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Cryptography Assignment - 5.

Theory - HW.

1. Rail fence cipher.

1(a) Encrypt "Enjoy your journey" using depth 2.

Given:

plaintext = ENJOY YOUR JOURNEY

Remove space

= ENJOY4OURJOURNEY

For depth 2.

Rail 1: E J Y O R O R E

Rail 2: N O Y U J U N Y

Rail 1: EJYORORE

Rail 2: NOYUJUNY

Ciphertext: EJYORORENOYUJUNY

1(b) Encrypt "ENJOY YOUR JOURNEY"

plaintext = ENJOY YOUR JOURNEY

Rail 1: E Y R R

Rail 2: N O Y U J U N Y

Rail 3: J O O E

Rail 1 = EYRR

Rail 2 = NOYUJUNY

Rail 3 = JOOE

Ciphertext = EYRRNOYUJUNYJOOE

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2. RAIL FENCE - MESSAGE OF MY CHOICE
plain text = HELLOWORLD

Depth 2.

write in two rails:

Rail 1: H L O O L

Rail 2: E L W R D

Rail 1 = HLOOL

Rail 2 = ELWRD

Ciphertext: HLOOLELWRD

11/3 - 3

Depth 3.

Rail 1 H O L

Rail 2 E L W R D

Rail 3 L O

Therefore.

Ciphertext: HOLELWRDLO

3. RAIL FENCE DECRYPTION.

Decrypt "EOCSNYUWLEJYREASONS"

E O C S

N Y U W L E

J Y R E A S

O N S

=> E N J O Y YOUR NEW CLASSES.

plain text: ENJOY YOUR NEW CLASSES.

4. RAIL FENCE DECRYPTION

Decrypt BOOIGUYUBOS NUUTYRKAS.
Depth 3.

Rail 1 B O O I G X
Rail 2 Y U B O S N U U T
Rail 3

Rail 1 B O O I G X
Rail 2 U Y U B O S N U U T
Rail 3 Y R K A S

plaintext: BUY YOUR BOOKS IN AUGUST

5. RAIL FENCE DECRYPTION

Decrypt SEMSERTISINCMOO using depth 2.
no of characters = 15.

Rail 1 S E M S E R T I
Rail 2 S I N C M O O

decryption string is

SEIMS NCEM ROT I

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6. Permutation cipher.
 1. Encrypt "We are discovered" using key 53124

5	3	1	2	4
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Cipher text : A S R R C E E I E E O D W D V

7. DES.

using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as:

Hexadecimal: 0x 0002 000 0000 0001.

Binary.

0000	0000	0000	0010
0000	0000	0000	0000
0000	0000	0000	0000
0000	0000	0000	0001

input to box
(plain text).

↓ 15th.

0000 0000 0000 0010 0000 0000 0000 0000
 0000 0000 0000 0000 0000 0000 0000 0010

Input bit 64 = 1, output position 25 = 1
 Input bit 15 = 1, output position 63 = 1.

By seeing permutation table.

64-bit output.

0000 0000 0000 0000 0000 0000 0000 0000
 0000 0000 0000 0000 0010 0000 0000 0000

output in hexadecimal 0x0000008000000002

- 2> Given the 4-bit plaintext (M) 1101 & Key (K) 1011. Calculate the resulting ciphertext (C) using a bitwise Exclusive-OR (XOR) operation. Show bit-by-bit work.

Solⁿ:

		XOR
	A	B
plaintext : 1101	1	1
Key : 1011	0	0
	0	1
	1	0
	1	1
	0	0

ciphertext: 0110

3. The input to S-box 1 is 000 000 what is the output.

Solⁿ

000 000 000 1st & 6th bit → row.

middle 4 bits → column.

1st 6th bit

So, 0th Row & 0th column see the number 14 in S-box → 14//.

Binary conversion of 14 → 1110.

output = 1110//.